

**Untrusted Input Ingress (Prompts,
Web Corpora, Documents, MCP Tools)**

Raw Ingress

Tier 1: Pre-Ingress ($\mathcal{K}_{\text{data}}$)
Canonicalize · Strip Macros
TrustTrace Provenance

[Interdicts]

- × RAG poisoning
- × Script macros
- × Cross-tenant leaks

Canonical Text

Tier 2: Model Guardrails ($\mathcal{K}_{\text{model}}$)
StruQ Typed Delimiters
SecAlign DPO · Sentinels

[Interdicts]

- × Indirect injection
- × Delimiter collision
- × Multi-turn jailbreaks

Aligned Context

Tier 3: Planning ($\mathcal{K}_{\text{CBAC}}$)
Ephemeral Scoped Tokens
Plan Critic · Invariants

[Interdicts]

- × Privilege pivoting
- × Deceptive plans
- × Invariant breaches

Revoke

Scoped Action

Tier 4: OS Sandboxing ($\mathcal{K}_{\text{sandbox}}$)
Micro-VM (gVisor/Kata)
Seccomp Filter · Deny Egress

[Interdicts]

- × Host shell / RCE
- × Namespace escapes
- × Outbound C2 exfil

Kill

Audit Events

Tier 5: Telemetry ($\mathcal{K}_{\text{telemetry}}$)
OORAR Drift · WORM Log
Circuit Breaker & Kill-Switch

[Interdicts]

- × Denial-of-Cash loops
- × Tool recursion
- × Byzantine drift

Kill-Switch

Request Execution Flow